SOURCE: MSC.197(80).pdf
FIGURE
1
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 102-
ANNEX 12
CRITERIA FOR LONGITUDINAL STRENGTH OF HULL GIRDER
FOR OIL TANKERS
1 General
1.1 These criteria should be used for the evaluation of the longitudinal strength of the ship's hull girder as required by 8.1.2.
1.2 In order that the ship's longitudinal strength to be evaluated can be recognized as valid, fillet welding between longitudinal internal members and hull envelopes should be in sound condition so as to keep the integrity of longitudinal internal members with hull envelopes.
2 Evaluation oflongitudinal strength
On oil tankers of 130 m in length and upwards and over 10 years of age, the longitudinal strength of the ship's hull girder should be evaluated in compliauce with the requirements of this annex on the basis of the thickness measured, renewed or reinforced, as appropriate, during the renewal survey of the Cargo Ship Safety Construction Certificate

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of this annex on the basis of the thickness measured, renewed or reinforced, as appropriate, during the renewal survey of the Cargo Ship Safety Construction Certificate or Cargo Ship Safety Certificate (SC renewal survey). The condition of the hull girder for longitudinal strength evaluation should be determined in accordance with the methods specified in appendix 3.
2.1 Calculation of transverse sectional areas of deck and bottom flanges of hull girder
2.1.1 The transverse sectional areas of deck flange (deck plating and deck longitudinals) and bottom flange (bottom shell plating and bottom longitudinals) of the ship's hull girder should be calculated by using the thickness measured, renewed or reinforced, as appropriate, during the SC renewal survey.
2.1.2 If the diminution of sectional areas of either deck or bottom flange exceeds 10% of their respective as-built area (i.e., original sectional area when the ship was built), either one ofthe following measures should be taken:
.1 to renew or reinforce the deck or bottom flanges so that the actual sectional area is not less than 90% of the as-built area; or

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either one ofthe following measures should be taken:
.1 to renew or reinforce the deck or bottom flanges so that the actual sectional area is not less than 90% of the as-built area; or
.2 to calculate the actual section of moduli (Zact) of transverse section of the ship's hull girder by applying the calculation method specified in appendix I, by using the thickness measured, renewed or reinforced, as appropriate, during the SC renewal survey.

2.2 Requirements for transverse section modulus of hull girder
The actual section moduli of the transverse section of the ship's hull girder, calculated in accordance with 2.1.2.2, should satisfy either of the following provisions, as applicable:
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.1 for ships constructed on or after 1 July 2002, the actual section moduli (Zaet) of the transverse section of the ship's hull girder calculated in accordance with the requirements of 2.1.2.2 should be not less than the

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.1 for ships constructed on or after 1 July 2002, the actual section moduli (Zaet) of the transverse section of the ship's hull girder calculated in accordance with the requirements of 2.1.2.2 should be not less than the diminution limits determined by the Administration, taking into account the recommendations adopted by the Organization; or
.2 for ships constructed before 1 July 2002, the actual section moduli (Zoet) of the transverse section of the ship's hull girder calculated in accordance with the requirements of 2.1.2.2 should meet the criteria for minimum section modulus for ships in service required by the Administration or recognized classification society, provided that in no case Zaet should be less than the diminution limit of the minimum section modulus (Zme) as specified in appendix 2.
Appendix 1
Calculation criteria of section moduli of midship section of hull girder
When calculating the transverse section modulus of the ship's hull girder, the sectional area of all continuous longitudinal strength members should be taken into account.
2 Large openings, i.e. openings exceeding 2.5 m in length or 1.2 m in breadth, and

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sectional area of all continuous longitudinal strength members should be taken into account.
2 Large openings, i.e. openings exceeding 2.5 m in length or 1.2 m in breadth, and scallops, where scallop welding is applied, are always to be deducted from the sectional areas used in the section modulus calculation.
3 Smaller openings (manholes, lightening holes, single scallops in way of seams, etc.) need not be deducted, provided that the sum of their breadths or shadow area breadths in one transverse section does not reduce the section modulus at deck or bottom by more than 3% and provided that he height of lightening holes, draining holes and single scallops in longitudinals or longitudinal girders does not exceed 25% of the web depth, for scallops maximum 75 mm.
4 A deduction-free sum of smaller opening breadths in one transverse section in the bottom or deck area of 0.06 (8 - Lb) (where 8 = breadth of ship, Lb = total breadth of large openings) may be considered equivalent to the above reduction in sectional modulus.
5 The shadow area will be obtained by drawing two tangent lines with an opening angle of30°.

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large openings) may be considered equivalent to the above reduction in sectional modulus.
5 The shadow area will be obtained by drawing two tangent lines with an opening angle of30°.
6 The deck modulus is related to the moulded deck line at side.
7 The bottom modulus is related to the baseline.
8 Continuous trunks and longitudinal hatch coamings should be included in the longitudinal sectional area provided they are effectively supported by longitudinal bulkheads or deep girders. The deck modulus is then to be calculated by dividing the moment of inertia by the following distance, provided this is greater than the distance to the deck line at side:
y, =y (0.9+0.2 ~)
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where:
y = distance from neutral axis to top of continuous strength member;
x = distance from top of continuous strength member to centreline of the ship;
x and y to be measured to the point giving the largest value of Yt .

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SOURCE: MSC.197(80).pdf
where:
y = distance from neutral axis to top of continuous strength member;
x = distance from top of continuous strength member to centreline of the ship;
x and y to be measured to the point giving the largest value of Yt .
9 Longitudinal girders between multi-hatchways will be considered by special calculations.
Appendix 2
Diminution limit of minimum longitudinal strength of ships in service
1 The diminution limit of the minimum section modulus (Zm,) of oil tankers in service is given by the following formula:
where:
L = Length of ship. L is the distance, in metres, on the summer load waterline from the fore side of stem to the after side of the rudder post, or the centre of the rudder stock if there is no rudder post. L should not be less than 96%, and need not be greater than 97%, of the extreme length on the summer load waterline. In ships with unusual stem and bow arrangement, the length L may be specially considered.
B = Greatest moulded breadth in metres.
Cb = Moulded block coefficient at draught d corresponding to summer load waterline , based on Land B. Cb should not be taken less than 0.6.
Cb = moulded displacement(m 3) at draught d LBd
C = 0.9cn

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B = Greatest moulded breadth in metres.
Cb = Moulded block coefficient at draught d corresponding to summer load waterline , based on Land B. Cb should not be taken less than 0.6.
Cb = moulded displacement(m 3) at draught d LBd
C = 0.9cn
Cn 10.75 - (300-L f 100 for 130 m::: L::: 300 m
Cn = 10.75 for 300 m < L < 350 m
Cn 10.75 - (L- 350 f 150 for 350 m::: L ::: 500 m
k = material factor, e.g.
k = 1.0 for mild steel with yield stress of 235 N/mm2 and over
k = 0.78 for high-tensile steel with yield stress of315 N/mm2 and over
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k = 0.72 for high-tensile steel with yield stress of355 N/mm 2 and over.
2 Scantlings of all continuous longitudinal members of the ship's hull girder based on the section modulus requirement in I above should be maintained within O.4L amidships. However, in special cases, based on consideration of type of ship, hull form and loading conditions, the scantlings may be gradually reduced towards the end of O.4L